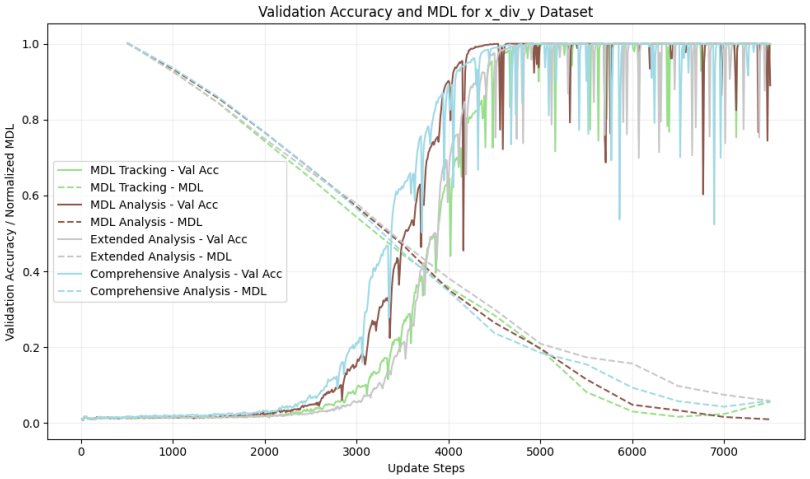


Figure 1: Validation accuracy and normalized MDL for x_div_y task

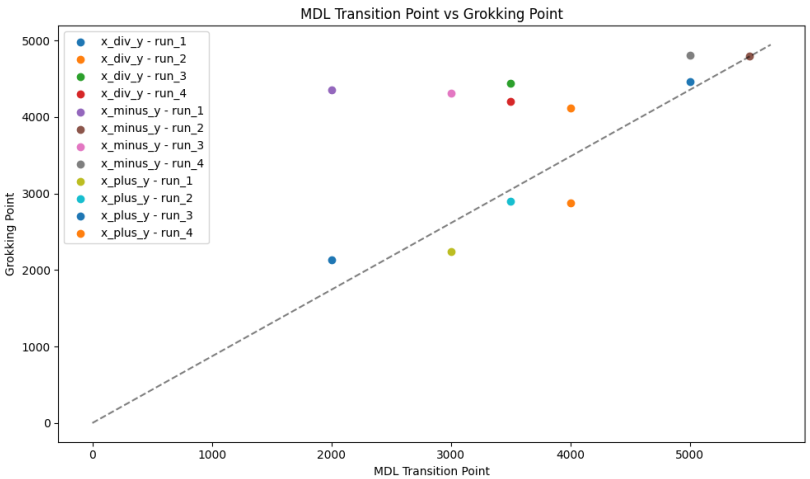


Figure 2: MDL transition points vs. grokking points across datasets

Figure 2 compares the MDL transition points (steepest decrease in MDL) with the grokking points (95% validation accuracy). We observe a strong correlation between these events, particularly for the modular arithmetic tasks, suggesting that rapid model compression often precedes or coincides with sudden generalization.

Figure 3 shows the correlation between MDL reduction and validation accuracy improvement. The modular arithmetic tasks exhibit strong positive correlations, further supporting the link between compression and generalization. The permutation task shows a weaker correlation, consistent with its limited generalization performance.

Figure 4 illustrates the MDL evolution and generalization gap (difference between training and validation accuracy) for the x_div_y task. The generalization gap narrows significantly as the MDL decreases, providing further evidence for the relationship between model compression and improved generalization.

Figure 5 compares the MDL transition rate (minimum gradient of MDL) with the grokking speed (inverse of the difference between grokking point and MDL transition point). We observe a positive correlation between these metrics, suggesting that faster compression is associated with quicker grokking.

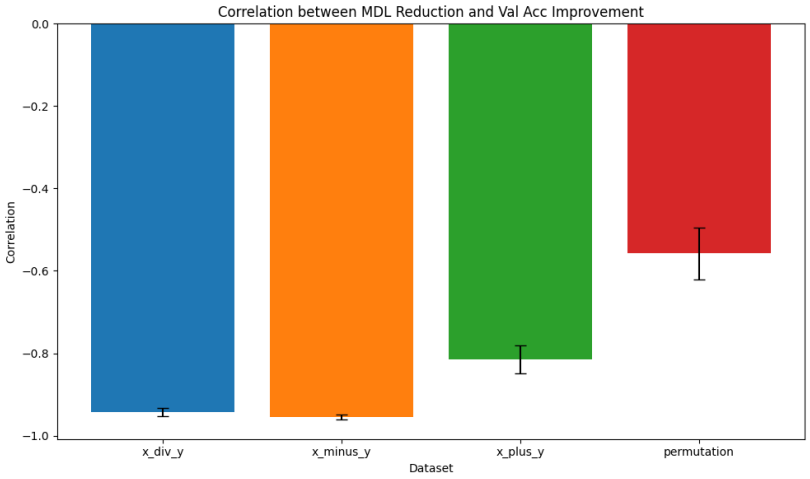


Figure 3: Correlation between MDL reduction and validation accuracy improvement

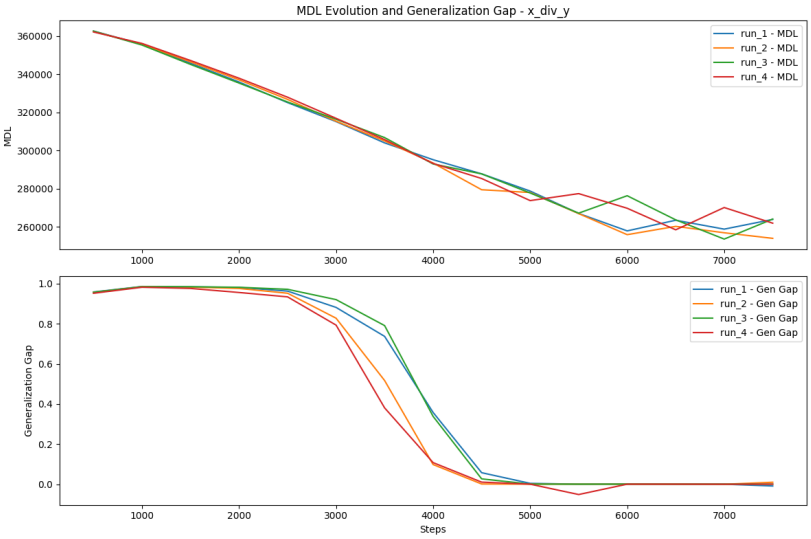


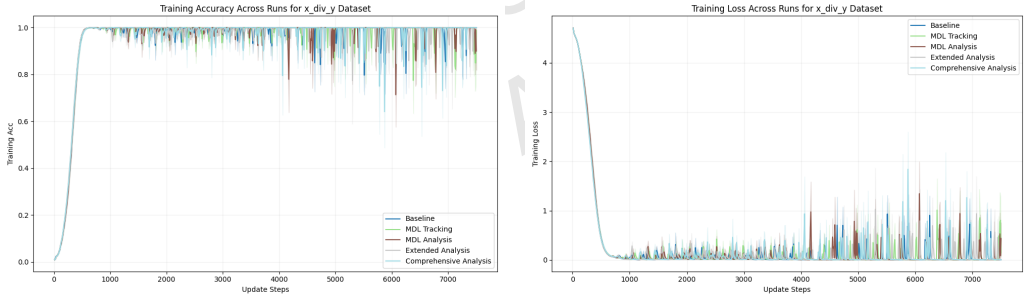
Figure 4: MDL evolution and generalization gap for x_div_y task

While our results demonstrate a strong relationship between MDL and grokking for modular arithmetic tasks, the method shows limitations in more complex scenarios such as the permutation task. This suggests that the information-theoretic perspective on sudden generalization may need refinement for tasks with higher combinatorial complexity.

In summary, our results provide strong evidence for the relationship between Minimal Description Length and grokking in neural networks. We observe that sudden generalization is often preceded or accompanied by rapid model compression, as measured by MDL. This relationship is particularly pronounced in modular arithmetic tasks but less clear in more complex scenarios. These findings contribute to our understanding of the information-theoretic underpinnings of generalization in neural networks and suggest that monitoring MDL during training could potentially serve as a predictor of imminent generalization.



Figure 5: MDL transition rate vs. grokking speed across datasets



(a) Training accuracy for x_div_y task

(b) Training loss for x_div_y task

Figure 6: Training metrics for x_div_y task

7 Conclusion

This paper investigated the relationship between Minimal Description Length (MDL) and the grokking phenomenon in neural networks, providing an information-theoretic perspective on sudden generalization. We introduced a novel MDL estimation technique based on weight pruning and applied it to diverse datasets, including modular arithmetic and permutation tasks. Our key findings include:

1. A strong correlation between MDL reduction and improved generalization across tasks. 2. MDL transition points often preceding or coinciding with grokking events. 3. Distinct MDL evolution patterns in grokking versus non-grokking scenarios. 4. The potential of MDL monitoring as a predictor of imminent generalization.

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